

Review of Mathematical background

Mathematical background

□ For any real number x ,

- $\lfloor x \rfloor$ (read “floor of x ”) is the largest integer less than or equal to x .
- $\lfloor 2.9 \rfloor = 2$
- $\lceil x \rceil$ (read “ceiling of x ”) is the smallest integer greater than or equal to x .
- $\lceil 6.1 \rceil = 7$

□ Modular arithmetic

- For any integer a and any positive integer n , the value $a \bmod n$ is the remainder (or residue) of the quotient a/n .
- $a \bmod n = a - n\lfloor a/n \rfloor$
- It follows that $0 \leq a \bmod n < n$
- If $(a \bmod n) = (b \bmod n)$, we write $a \equiv b \pmod{n}$ and say that a is equivalent to b , modulo n .

Exponentials

□ For all real $a > 0$, m , and n , we have the following:

- $a^0 = 1$, $a^1 = a$, $a^{-1} = 1/a$
- $(a^m)^n = a^{mn} = (a^n)^m$

□ The function a^n is monotonically increasing with n .

□ Using e to denote 2.71828..., the base of the natural logarithm function, we have

- $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{i=0}^{\infty} \frac{x^i}{i!}$, where $!$ is the factorial function
- We also have: $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$

Logarithms

- ❑ For $b > 1$ and $x > 0$, $\log_b x$ (read “log to the base b of x ”) is that number y such that $b^y = x$
- ❑ $\log_b$ is a strictly increasing function, i.e., if $x_1 > x_2$, then $\log_b x_1 > \log_b x_2$
- ❑ $\log_b 1 = 0$
- ❑ $\log_c(ab) = \log_c a + \log_c b$
- ❑ $\log_b(x^a) = a \log_b x$

- ❑ $\lg x = \log_2 x$, binary logarithm
- ❑ $\ln x = \log_e x$, natural logarithm

- ❑ $\log_b a = \frac{\log_c a}{\log_c b}$
- ❑ $a^{\log_b c} = c^{\log_b a}$

Factorials

□ Factorials

- For integers $n \geq 0$

$$n! = f(x) = \begin{cases} 1, & \text{if } n = 0 \\ n \cdot (n - 1)!, & \text{if } n > 0 \end{cases}$$

- Thus, $n! = 1 \cdot 2 \cdot 3 \cdot \dots (n - 1) \cdot n$

Permutations

- A permutation of a finite set S is an **ordered sequence** of all the elements of S , with each element appearing exactly once.
 - If $S = \{a, b, c\}$, then S has 6 permutations:
 - ✓ $abc, acb, bac, bca, cab, cba$
 - There are $n!$ permutations of a set of n elements
 - ✓ Choose the first element in n ways, the second in $n-1$ ways, the third in $n-2$ ways, and so on.
- A **k -permutation** of S is an **ordered sequence of k elements of S** , with no element appearing more than once.
 - If $S = \{a, b, c\}$, then S has six 2-permutations
 - ✓ ab, ba, ac, ca, bc, cb
 - The number of k -permutations of a set of n elements: ${}_nP_k = n(n-1)(n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}$
 - ✓ Choose the first element in n ways, the second in $n-1$ ways, ..., and the k -th element in $n-k+1$ ways

Combinations

- ❑ Choosing k **distinct** elements from S , and the **order does not matter**.
 - The 4-set $S = \{a, b, c, d\}$ has six 2-combinations: ab, ac, ad, bc, bd, cd
- ❑ Related to k -permutation
- ❑ Every k -combination has exactly $k!$ permutations of its elements, each of which is a distinct k -permutation of S .
- ❑ Thus, the number of k -combination is the number of k -permutations divided by $k!$

Write as ${}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$, read as “ n choose k ”

Symmetric: $\binom{n}{k} = \binom{n}{n-k}$

Summation formulas 1

□ Arithmetic series

- $\sum_{k=1}^n k = 1 + 2 + \cdots + n = \frac{1}{2}n(n+1)$

□ Sums of squares and cubes

- $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6},$
- $\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$

□ Geometric series

- $\sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$, for real $x \neq 1$
- $\sum_{k=0}^n x^k = \frac{x^{n+1}-1}{x-1}$
- $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$, when $|x| < 1$

Summation formulas 2

□ Harmonic series

- $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} = \sum_{k=1}^n \frac{1}{k} = \ln n + \text{Some constant}$

□ Integrating and differentiating series

- E.g., differentiating both sides of the infinite geometric series, and multiply by x:
- $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x} \rightarrow \sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2}$